

Maharashtra Public Service Commission Assessment

Question Paper Name :	CHROMA1K Biology QP2
Subject Name :	Assistant Professor in Biology Group A and Lecturer in Biology Group B
Creation Date :	2026-07-22 11:04:20
Duration :	60
Number of Questions :	100
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes

Assistant Professor in Biology Group A and Lecturer in Biology Group B

Group Number :	1
Group Id :	4410092963469
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0

Assistant Professor in Biology Group A and Lecturer in Biology Group B

Section Id :	4410092993154
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100
Number of Questions to be attempted :	100
Maximum Instruction Time :	0

Sub-Section Number : 1
Sub-Section Id : 4410093004125
Question Shuffling Allowed : Yes
Is Section Default? : No

Question Number : 1 Question Id : 4410092112691 Question Type : MCQ Option Shuffling : No

Which of the following best describes a hydrophobic interaction in biological molecules?

Options :

1. Attraction between non-polar molecules in water
2. Bond formation between two sulfur atoms in proteins
3. Hydrogen bonding between polar groups
4. Electrostatic attraction between charged ions

Question Number : 2 Question Id : 4410092112694 Question Type : MCQ Option Shuffling : No

Which colligative property is used to determine the molar mass of a solute in solution?

Options :

1. Osmosis
2. Absorbance
3. Conductivity
4. Viscosity

Question Number : 3 Question Id : 4410092112697 Question Type : MCQ Option Shuffling : No

Which property allows an enzyme to distinguish between similar molecules during catalysis?

Options :

1. Stability
2. Affinity
3. Activity
4. Specificity

Question Number : 4 Question Id : 4410092112702 Question Type : MCQ Option Shuffling : No

Which enzyme is essential for processing primary microRNA into precursor microRNA in the nucleus?

Options :

1. Dicer
2. Drosha
3. Ligase
4. Polymerase

Question Number : 5 Question Id : 4410092112687 Question Type : MCQ Option Shuffling : No

Which nitrogenous base pairs with guanine in DNA?

Options :

1. Uracil
2. Cytosine
3. Adenine
4. Thymine

Question Number : 6 Question Id : 4410092112689 Question Type : MCQ Option Shuffling : No

Which class of biomolecules is crucial for catalyzing metabolic reactions in cells?

Options :

1. Lipid
2. Nucleic acid
3. Carbohydrate
4. Proteins

Question Number : 7 Question Id : 4410092112701 Question Type : MCQ Option Shuffling : No

Which unusual base is commonly found in the D loop of tRNA?

Options :

1. Pseudouridine
2. Methylguanosine
3. Inosine
4. Dihydrouridine

Question Number : 8 Question Id : 4410092112705 Question Type : MCQ Option Shuffling : No

Which enzyme is primarily responsible for converting inosine monophosphate (IMP) to adenylosuccinate during purine nucleotide biosynthesis?

Options :

1. IMP dehydrogenase
2. PRPP synthetase
3. Adenylate kinase
4. Adenylosuccinate synthetase

Question Number : 9 Question Id : 4410092112706 Question Type : MCQ Option Shuffling : No

What is the arrangement of hydrophobic and hydrophilic regions in a lipid bilayer?

Options :

1. Neutral
2. Amphipathic
3. Nonpolar
4. Isotonic

Question Number : 10 Question Id : 4410092112715 Question Type : MCQ Option Shuffling : No

Which enzyme is uniquely associated with peroxisomes and plays a central role in hydrogen peroxide metabolism?

Options :

1. Oxidase
2. Dehydrogenase
3. Catalase
4. Kinase

Question Number : 11 Question Id : 4410092112716 Question Type : MCQ Option Shuffling : No

Why do storage vacuoles in seeds differ from those in leaf cells?

Options :

1. They synthesize chlorophyll for energy capture
2. They accumulate proteins or oils needed for germination
3. They increase photosynthetic activity in seeds
4. They facilitate gas exchange during development

Question Number : 12 Question Id : 4410092112720 Question Type : MCQ Option Shuffling : No

Which statement best explains why interrupted genes are considered advantageous in eukaryotic genomes?

Options :

1. They restrict gene expression to certain tissue types
2. They ensure consistently high rates of transcription
3. They provide flexibility in gene expression through exon shuffling
4. They maintain unchanged gene sequences across generations

Question Number : 13 Question Id : 4410092112714 Question Type : MCQ Option Shuffling : No

If a mutation prevents the formation of Golgi-derived vesicles, what is the most direct effect on plant cell physiology?

Options :

1. DNA replication is disrupted
2. Secretion of cell wall components is inhibited
3. Mitochondrial respiration is reduced
4. Protein synthesis in the cytoplasm is halted

Question Number : 14 Question Id : 4410092112718 Question Type : MCQ Option Shuffling : No

Which cellular component regulates protrusion at the leading edge of migrating cells?

Options :

1. Centrosome
2. Filopodium
3. Lamellipodium
4. Lysosome

Question Number : 15 Question Id : 4410092112719 Question Type : MCQ Option Shuffling : No

How would a mutation within a unique DNA segment most likely affect an organism compared to a mutation in highly repetitive DNA?

Options :

1. Mutation in unique DNA is less likely to affect cellular processes
2. Mutation in unique DNA can directly alter gene function and phenotype
3. Mutation in unique DNA is more difficult to detect experimentally
4. Mutation in unique DNA increases overall genome size

Question Number : 16 Question Id : 4410092112721 Question Type : MCQ Option Shuffling : No

A researcher is studying the evolutionary relationship between several gene families found on different chromosomes in a plant species. Which method would most effectively help the researcher determine whether these gene families originated from a common ancestral gene?

Options :

1. Comparative sequence analysis of gene family members
2. Expression analysis of gene family members
3. Chromosomal staining for gene localization
4. Protein localization analysis of gene family members

Question Number : 17 Question Id : 4410092112897 Question Type : MCQ Option Shuffling : No

Which mode of DNA replication is observed in all organisms?

Options :

1. Random
2. Semiconservative
3. Dispersive
4. Conservative

Question Number : 18 Question Id : 4410092112902 Question Type : MCQ Option Shuffling : No

Which process is responsible for repairing thymine-thymine dimers caused by UV radiation in DNA?

Options :

1. Base excision repair
2. Recombination repair
3. Nucleotide excision repair
4. Mismatch repair

Question Number : 19 Question Id : 4410092112901 Question Type : MCQ Option Shuffling : No

DNA damage triggered inactivation of Cdc25 type protein phosphatases arrests the cell cycle primarily at which of the following steps?

Options :

1. Chromosome condensation
2. Entry into mitosis
3. G1 / S phase transition
4. DNA replication inhibition

Question Number : 20 Question Id : 4410092112908 Question Type : MCQ Option Shuffling : No

What is the main function of the 5' cap present on an mRNA molecule?

Options :

1. Facilitating codon-anticodon pairing during translation
2. Protecting mRNA from degradation by nucleases
3. Catalyzing transcription by RNA polymerase II
4. Joining exons during RNA splicing

Question Number : 21 Question Id : 4410092112912 Question Type : MCQ Option Shuffling : No

Where do the large and small ribosomal subunits combine to form a functional ribosome during translation?

Options :

1. Endoplasmic reticulum
2. Cytoplasm
3. Golgi body
4. Vacuole

Question Number : 22 Question Id : 4410092112905 Question Type : MCQ Option Shuffling : No

Which of the following enzymes is coded by the genome of retrovirus?

Options :

1. RNA polymerase II
2. ATP synthase
3. Reverse transcriptase
4. Peptidyl transferase

Question Number : 23 Question Id : 4410092112910 Question Type : MCQ Option Shuffling : No

In certain RNA editing systems, which class of RNAs supplies the specific sequence information necessary to direct nucleotide changes in the RNA transcript?

Options :

1. Ribosomal RNAs (rRNAs)
2. Small nuclear RNAs (snRNAs)
3. Guide RNAs (gRNAs)
4. Transfer RNAs (tRNAs)

Question Number : 24 Question Id : 4410092112913 Question Type : MCQ Option Shuffling : No

Which factor is responsible for proofreading of the codon–anticodon interaction during elongation phase of protein synthesis in prokaryotes?

Options :

1. Elongation factor Tu
2. Elongation factor Ts
3. Elongation factor 2
4. Elongation factor G

Question Number : 25 Question Id : 4410092112920 Question Type : MCQ Option Shuffling : No

What will happen when a cell is exposed to a histone deacetylase (HDAC) inhibitor?

Options :

1. Transcription of some genes will increase
2. Chromatin condensation will occur
3. Histone variants will be inserted into nucleosomes
4. DNA will be exposed to nucleases

Question Number : 26 Question Id : 4410092112921 Question Type : MCQ Option Shuffling : No

Which class of small RNA molecules is commonly involved in RNA interference-mediated gene silencing in eukaryotes?

Options :

1. tRNA
2. rRNA
3. miRNA
4. snRNA

Question Number : 27 Question Id : 4410092112924 Question Type : MCQ Option Shuffling : No

Which pathogen is responsible for causing fire blight disease in pear and apple trees?

Options :

1. *Erwinia amylovora*
2. *Agrobacterium tumefaciens*
3. *Xanthomonas oryzae*
4. *Puccinia graminis*

Question Number : 28 Question Id : 4410092112923 Question Type : MCQ Option Shuffling : No

What happens to soluble nitrogen content in plant cells during rapid viral synthesis caused by virus infection?

Options :

1. It decreases
2. It remains unchanged
3. It is permanently stored in vacuoles
4. It converts into carbohydrates

Question Number : 29 Question Id : 4410092112928 Question Type : MCQ Option Shuffling : No

Which process initiates cell surface receptor-mediated signaling in bacteria?

Options :

1. Ligand-induced activation of receptor tyrosine kinase
2. Steroid hormone binding to its receptor
3. G-protein-coupled receptor activation
4. Ligand binding to a sensor histidine kinase

Question Number : 30 Question Id : 4410092112941 Question Type : MCQ Option Shuffling : No

Which condition is most frequently linked to inherited mutations in BRCA1 and BRCA2 genes?

Options :

1. Infectious viral diseases
2. Lung and liver diseases only
3. Breast and ovarian cancers
4. Neurological disorders only

Question Number : 31 Question Id : 4410092112929 Question Type : MCQ Option Shuffling : No

Where are the receptors that initiate signaling by brassinosteroids located in a plant cell?

Options :

1. Nucleus
2. Cytoplasm
3. Plasma membrane
4. Endoplasmic reticulum

Question Number : 32 Question Id : 4410092112935 Question Type : MCQ Option Shuffling : No

If $\alpha 4\beta 1$ integrin expression is reduced on hematopoietic cells, what is the most likely outcome?

Options :

1. Impaired attachment to fibronectin and reduced differentiation
2. Increased deposition of calcium in bone matrix
3. Enhanced secretion of extracellular matrix proteins
4. Reduced hemoglobin synthesis in circulating erythrocytes

Question Number : 33 Question Id : 4410092112937 Question Type : MCQ Option Shuffling : No

The loss of pectin-mediated adhesion in meristematic cells can result in:

Options :

1. Arrest of meristematic cell division
2. Formation of new shoots from the meristematic cells
3. Loose cell association with impaired differentiation
4. Thickening of meristematic cell walls

Question Number : 34 Question Id : 4410092112938 Question Type : MCQ Option Shuffling : No

Which combination of statements clearly reflects the roles and distribution of the Extracellular Matrix (ECM)?

1. It participates in cell adhesion
2. It stores signaling molecules
3. It is present only in plants
4. It influences cell differentiation

Options :

1. 1, 2, 3 and 4
2. 1 and 3 only
3. 1, 2 and 4 only
4. 2 and 3 only

Question Number : 35 Question Id : 4410092112944 Question Type : MCQ Option Shuffling : No

What important role carotenoids play besides serving as accessory light harvesting pigments in plants?

Options :

1. CO₂ fixation
2. Photoprotection
3. NADPH synthesis
4. Oxygen evolution

Question Number : 36 Question Id : 4410092112950 Question Type : MCQ Option Shuffling : No

Which ammonium assimilation pathway produces glutamate directly by utilizing NADH or NADPH as a reductant?

Options :

1. Glutamine synthase - glutamine 2-oxoglutarate aminotransferase pathway
2. Glutamate dehydrogenase pathway
3. Glycolate - Glyoxylate pathway
4. Nitrate reductase - Nitrite reductase pathway

Question Number : 37 Question Id : 4410092112955 Question Type : MCQ Option Shuffling : No

Type I phytochrome is encoded by _____ and is _____.

Options :

1. PHYB gene; photoreversible
2. PHYC gene; thermosensitive
3. PHYA gene; photolabile
4. PHYE gene; thermostable

Question Number : 38 Question Id : 4410092112951 Question Type : MCQ Option Shuffling : No

In a C₄ plant exposed to ¹⁴CO₂ under illumination, which compound is expected to show the earliest incorporation of radioactive carbon?

Options :

1. Glycine
2. Phosphoglycerate
3. Oxaloacetate
4. Glycolate

Question Number : 39 Question Id : 4410092112954 Question Type : MCQ Option Shuffling : No

Which of the following statements related to plant hormones can be considered as a characteristic of climacteric fruit?

Options :

1. Increase in gibberellin levels
2. Increase in auxin levels
3. Increase in levels of abscisic acid
4. Increase in endogenous ethylene levels

Question Number : 40 Question Id : 4410092112957 Question Type : MCQ Option Shuffling : No

How stomatal movements get affected if proton pumping from the cytosol of guard cells to the apoplast is inhibited?

Options :

1. ABA levels in guard cells will decrease, causing stomatal closure
2. Influx of K^+ ions into guard cells will decrease, causing stomatal closure
3. Guard cell cytoplasm will be acidified, causing stomatal opening
4. Water transport into guard cells will increase, causing stomatal opening

Question Number : 41 Question Id : 4410092112961 Question Type : MCQ Option Shuffling : No

A farmer applies a plant-derived pyrethrin extract to his crop. What effect will this application have on his crop?

Options :

1. It will promote growth of the crop
2. It will kill insects pests on the crop
3. It will act as a nitrogen fertilizer to the crop
4. It will protect the crop against drought stress

Question Number : 42 Question Id : 4410092113136 Question Type : MCQ Option Shuffling : No

Which notation custom is commonly used to represent wild-type alleles in genetics?

Options :

1. A superscript plus sign after the gene letter
2. No symbol, just the gene letter
3. A capital letter instead of lowercase
4. A minus sign after the gene letter

Question Number : 43 Question Id : 4410092113139 Question Type : MCQ Option Shuffling : No

What is the molecular mark that conditions the expression of an imprinted gene in mammals?

Options :

1. Methylation of CpG dinucleotides near the gene
2. Phosphorylation of histone proteins
3. Ubiquitination of transcription factors
4. Acetylation of promoter regions

Question Number : 44 Question Id : 4410092113141 Question Type : MCQ Option Shuffling : No

What does a recombinant frequency of less than 50 per cent between two genes indicate in a testcross?

Options :

1. Genes are non-homologous
2. Genes assort independently
3. Genes are unlinked
4. Genes are linked

Question Number : 45 Question Id : 4410092113145 Question Type : MCQ Option Shuffling : No

Why is specialized transduction limited to only certain bacterial genes?

Options :

1. It involves random selection during packaging
2. It depends on recombination frequency
3. It depends on proximity to the phage integration site
4. It is due to phage mutation

Question Number : 46 Question Id : 4410092113143 Question Type : MCQ Option Shuffling : No

If a flower develops on a variegated branch, what non-mendelian factor determines the phenotype of the progeny?

Options :

1. Distribution of chloroplasts in egg cells
2. Cytoplasmic mixing from pollen
3. Paternal genotype influence
4. Segregation of nuclear genes

Question Number : 47 Question Id : 4410092113146 Question Type : MCQ Option Shuffling : No

What makes sexduction distinct from standard $F^+ \times F^-$ conjugation?

Options :

1. Sexduction occurs at a slower rate than $F^+ \times F^-$ conjugation
2. Sexduction uses viral intermediates for gene transfer
3. Bacterial genes on the F' factor are transferred at higher frequency
4. Sexduction prevents transfer of bacterial genes

Question Number : 48 Question Id : 4410092113147 Question Type : MCQ Option Shuffling : No

In which scenario would a LOD score of -2.5 most likely be obtained?

Options :

1. The two loci are assorting independently and are not genetically linked.
2. There is inconclusive evidence about the linkage status.
3. The two loci are always inherited together.
4. The two loci are closely linked on the chromosome.

Question Number : 49 Question Id : 4410092113150 Question Type : MCQ Option Shuffling : No

What type of mutation is most likely to result if a DNA base exists in its rare tautomeric form during DNA replication?

Options :

1. Deamination
2. Base-pair substitution
3. Frameshift mutation
4. Exon skipping

Question Number : 50 Question Id : 4410092113159 Question Type : MCQ Option Shuffling : No

How does cell specialization contribute to the evolution of structural complexity in multicellular organisms?

Options :

1. It prevents cells from coordinating functions
2. It leads to division of labor and organized tissues or organs
3. It results in all cells being identical
4. It reduces the need for cellular cooperation

Question Number : 51 Question Id : 4410092113156 Question Type : MCQ Option Shuffling : No

When more than one scientific name has been assigned to the same taxon, which nomenclatural principle is used to determine the accepted name?

Options :

1. Principle of Typification
2. Binomial Principle
3. Principle of Priority
4. Principle of Uniqueness

Question Number : 52 Question Id : 4410092111114 Question Type : MCQ Option Shuffling : No

The Sundarbans are best known for which characteristic ecosystem?

Options :

1. Dry deciduous forests
2. Tropical grasslands
3. Alpine meadows
4. Mangrove forests

Question Number : 53 Question Id : 4410092111115 Question Type : MCQ Option Shuffling : No

Which region of the Indian subcontinent is a major centre of vertebrate endemism?

Options :

1. Indo-Gangetic Plain
2. Western Ghats
3. Deccan Plateau
4. Thar Desert

Question Number : 54 Question Id : 4410092111113 Question Type : MCQ Option Shuffling : No

State which of the following Options is correct regarding the Assertion and Reason given below.

Assertion (A): Homologous structures indicate evolutionary relationships.

Reason (R): Homologous structures perform identical functions in all organisms.

Options :

1. A is false, but R is true
2. Both A and R is true, but R is not the correct explanation of A
3. Both A and R is true, and R is the correct explanation of A
4. A is true, but R is false

Question Number : 55 Question Id : 4410092111118 Question Type : MCQ Option Shuffling : No

Which migratory bird arrives in the Indian subcontinent during the winter season?

Options :

1. Purple Sunbird
2. Rose-ringed Parakeet
3. Bar-headed Goose
4. Indian Peafowl

Question Number : 56 Question Id : 4410092113167 Question Type : MCQ Option Shuffling : No

Which trait is commonly found in r-selected species?

Options :

1. Early maturity and short lifespan
2. Delayed maturity and high parental care
3. Few offspring and long gestation
4. Large body size and stable environments

Question Number : 57 Question Id : 4410092113169 Question Type : MCQ Option Shuffling : No

What is a possible evolutionary outcome when populations remain geographically isolated over a long period?

Options :

1. Gene pool uniformity
2. Emergence of new species
3. Immediate reproductive isolation
4. Return to ancestral forms

Question Number : 58 Question Id : 4410092113173 Question Type : MCQ Option Shuffling : No

Which characteristic distinguishes a community from a population in ecology?

Options :

1. Same genetic makeup
2. Individuals of a single species
3. Presence of multiple interacting species
4. No interaction among organisms

Question Number : 59 Question Id : 4410092113181 Question Type : MCQ Option Shuffling : No

Which aquatic ecosystem is characterized by being the largest and most stable?

Options :

1. Deep sea ecosystem
2. Estuarine ecosystem
3. Riverine ecosystem
4. Freshwater ecosystem

Question Number : 60 Question Id : 4410092113163 Question Type : MCQ Option Shuffling : No

A botanist introduces a plant to a new habitat without competitors, predators, or diseases. Which term describes the range the plant could occupy?

Options :

1. Ecological optimum
2. Fundamental niche
3. Competitive range
4. Realized niche

Question Number : 61 Question Id : 4410092113166 Question Type : MCQ Option Shuffling : No

Character displacement helps to prevent which process among coexisting species?

Options :

1. Competitive exclusion
2. Resource partitioning
3. Reproductive isolation
4. Mutualism

Question Number : 62 Question Id : 4410092113168 Question Type : MCQ Option Shuffling : No

Which conditions support exponential (J-shaped) population growth?

Options :

1. Unlimited resources and no environmental resistance
2. Carrying capacity and population stability
3. Constant predation and resource scarcity
4. Limited resources and high competition

Question Number : 63 Question Id : 4410092113182 Question Type : MCQ Option Shuffling : No

Which combination best represents pollutants emitted from agricultural activities?

Options :

1. Sulfur dioxide and ash
2. Carbon monoxide and soot
3. Ammonia and methane
4. Nitrogen oxides and hydrocarbons

Question Number : 64 Question Id : 4410092113184 Question Type : MCQ Option Shuffling : No

Which zone of a biosphere reserve permits sustainable agriculture and settlements?

Options :

1. Exclusion zone
2. Core zone
3. Buffer zone
4. Transition zone

Question Number : 65 Question Id : 4410092113186 Question Type : MCQ Option Shuffling : No

Which example provides experimental proof for evolution by natural selection?

Options :

1. Pollination in flowering plants
2. Migration of birds in winter
3. Photosynthesis in green plants
4. Melanism in peppered moths

Question Number : 66 Question Id : 4410092113189 Question Type : MCQ Option Shuffling : No

What did the Miller-Urey experiment show about the primitive Earth conditions?

Options :

1. Protocells spontaneously assembled from polymers
2. Lipids were synthesized by volcanic activity
3. Nucleotides directly formed from clay minerals
4. Amino acids could form under simulated primitive conditions

Question Number : 67 Question Id : 4410092113191 Question Type : MCQ Option Shuffling : No

Which cellular structure is supposed to have originated from an engulfed aerobic bacterium according to the Endosymbiotic Theory?

Options :

1. Chloroplast
2. Mitochondrion
3. Endoplasmic reticulum
4. Nucleus

Question Number : 68 Question Id : 4410092113193 Question Type : MCQ Option Shuffling : No

Which evolutionary process can be prevented by gene flow occurring between populations?

Options :

1. Speciation
2. Natural selection
3. Increased rate of mutation
4. Extinction

Question Number : 69 Question Id : 4410092111124 Question Type : MCQ Option Shuffling : No

Choose the INCORRECT statement about the use of space in animal behavior.

Options :

1. Defence costs can limit territory size.
2. Territoriality guarantees access to every resource.
3. Individual home ranges may overlap spatially.
4. Territorial signals can reduce direct encounters.

Question Number : 70 Question Id : 4410092111123 Question Type : MCQ Option Shuffling : No

Which statement about social dominance and gene expression in the brain is correct?

Options :

1. Dominance only upregulates stress-response genes
2. Dominance silences all neural gene activity
3. Dominance alters expression of genes for synaptic plasticity
4. Dominance standardizes gene expression across individuals

Question Number : 71 Question Id : 4410092113370 Question Type : MCQ Option Shuffling : No

What is the primary objective of using transgenic plants for molecular farming?

Options :

1. Producing valuable pharmaceutical proteins using plant expression systems.
2. Producing modified plant structures using environmental growth conditions.
3. Producing improved soil nutrients using agricultural management systems.
4. Producing altered physical traits using traditional cultivation practices.

Question Number : 72 Question Id : 4410092113369 Question Type : MCQ Option Shuffling : No

Which agricultural application of tissue culture is most effective for recovering hybrids from incompatible crosses?

Options :

1. Meristem culture for pathogen-free plant production
2. Embryo culture for rescue of hybrid embryos
3. Cell suspension culture for biomass production
4. Somatic embryogenesis for clonal plant regeneration

Question Number : 73 Question Id : 4410092113372 Question Type : MCQ Option Shuffling : No

A genetically modified tomato variety was developed to increase post-harvest storage life by altering the activity of an enzyme involved in fruit maturation. Which transgenic modification was used to achieve this objective?

Options :

1. Suppression of polygalacturonase gene expression using antisense RNA technology.
2. Introduction of *Bacillus thuringiensis* genes producing insecticidal proteins.
3. Expression of carotenoid biosynthesis genes increasing β -carotene production.
4. Incorporation of herbicide-resistance genes allowing selective chemical tolerance.

Question Number : 74 Question Id : 4410092113371 Question Type : MCQ Option Shuffling : No

A recombinant therapeutic protein produced in plants shows the correct amino acid sequence and high expression level but differs in activity compared with the same protein produced in mammalian cells. Which analysis would directly address the cause of functional variation?

Options :

1. Evaluating post-translational modifications affecting final protein properties.
2. Evaluating promoter activity controlling total transcript accumulation levels.
3. Evaluating transformation efficiency controlling regenerated plant numbers.
4. Evaluating selectable markers controlling transformed tissue recovery rates.

Question Number : 75 Question Id : 4410092113373 Question Type : MCQ Option Shuffling : No

A crop engineered for insect protection shows excellent performance initially, but pest survival increases after several growing seasons. Which approach would provide a more sustainable solution?

Options :

1. Introducing traits with different biological targets to reduce adaptation pressure.
2. Increasing expression levels of the same trait throughout plant development.
3. Selecting plants showing maximum expression during early growth stages only.
4. Maintaining cultivation practices that continuously favour a single trait response.

Question Number : 76 Question Id : 4410092113379 Question Type : MCQ Option Shuffling : No

A plant exposed to cadmium-contaminated soil shows increased synthesis of small cysteine-rich peptides. Mutants unable to produce these molecules exhibit severe growth inhibition under the same conditions. Which function is primarily affected in these mutants?

Options :

1. Formation of stable complexes that reduce intracellular toxicity of metal ions.
2. Conversion of metal ions into organic molecules required for plant metabolism.
3. Enhancement of photosynthetic reactions through metal-dependent pathways.
4. Activation of reproductive processes during unfavorable growth conditions.

Question Number : 77 Question Id : 4410092113381 Question Type : MCQ Option Shuffling : No

Which factor is most important for preserving protein activity during the isolation process?

Options :

1. Maintaining the sample at low temperature
2. Increasing the incubation time of samples
3. Using highly concentrated salt solutions
4. Performing extraction under bright light

Question Number : 78 Question Id : 4410092113384 Question Type : MCQ Option Shuffling : No

Which feature distinguishes lentiviral vectors from conventional retroviral vectors in gene delivery?

Options :

1. They efficiently transduce both dividing and non-dividing cells
2. They replicate independently without entering the host nucleus
3. They package larger DNA fragments than adenoviral vectors
4. They express transgenes without requiring reverse transcription

Question Number : 79 Question Id : 4410092113386 Question Type : MCQ Option Shuffling : No

Which characteristic distinguishes a genomic library from a cDNA library prepared from the same organism?

Options :

1. It contains introns, promoters, and intergenic DNA sequences.
2. It contains only protein-coding regions expressed in all tissues.
3. It lacks repetitive DNA sequences present in the genome.
4. It represents only genes actively translated into proteins.

Question Number : 80 Question Id : 4410092113391 Question Type : MCQ Option Shuffling : No

A researcher measures purified DNA using a UV-Visible spectrophotometer and obtains an absorbance maximum at 260 nm. Which molecular property is mainly responsible for this absorption?

Options :

1. Electronic transitions occurring in aromatic nitrogenous bases of nucleic acids.
2. Vibrational transitions occurring in phosphate groups of nucleic acid molecules.
3. Rotational transitions occurring in sugar molecules of nucleotide structures.
4. Structural transitions occurring in hydrogen bonds between DNA strands.

Question Number : 81 Question Id : 4410092113394 Question Type : MCQ Option Shuffling : No

A scientist selects MALDI mass spectrometry for analyzing a purified protein sample. What is the main role of the matrix used in this technique?

Options :

1. It assists ion formation while protecting biomolecules from excess fragmentation.
2. It enhances protein folding while maintaining native biological activity completely.
3. It increases peptide separation while preventing interaction between molecules.
4. It directly determines molecular mass by detecting the flight time of generated ions.

Question Number : 82 Question Id : 4410092113385 Question Type : MCQ Option Shuffling : No

Phenol:chloroform extraction separates nucleic acids from proteins because proteins preferentially partition into the:

Options :

1. Organic phase after denaturation by phenol
2. Aqueous phase after hydration by buffer
3. Interphase due to increased ionic strength only
4. Pellet following centrifugation without denaturation

Question Number : 83 Question Id : 4410092113388 Question Type : MCQ Option Shuffling : No

Which experimental observation most strongly suggests that a cloning vector is a cosmid rather than a λ replacement vector?

Options :

1. Recombinant DNA is packaged into phage heads but propagates thereafter as plasmids.
2. Recombinant DNA integrates into bacterial chromosomes before plasmid replication begins.
3. Recombinant DNA produces mature bacteriophages following every bacterial infection cycle.
4. Recombinant DNA expresses viral structural proteins throughout bacterial propagation.

Question Number : 84 Question Id : 4410092113395 Question Type : MCQ Option Shuffling : No

A researcher analyzes the segregation of a dominant genetic marker in 80 offsprings, assuming a probability of 0.75 for the dominant phenotype. Which characteristic of the resulting binomial distribution is correct?

Options :

1. The distribution is negatively skewed because the probability of success exceeds one-half.
2. The distribution is positively skewed because the probability of success exceeds one-half.
3. The distribution is perfectly symmetric because the number of trials is sufficiently large.
4. The distribution is uniformly distributed because the probability remains constant.

Question Number : 85 Question Id : 4410092113397 Question Type : MCQ Option Shuffling : No

Two regression models are developed using the same biotechnology dataset. Model I has an R^2 of 0.82, whereas Model II has an R^2 of 0.56. Assuming both models satisfy all regression assumptions, which conclusion is most appropriate?

Options :

1. Model I explains a greater proportion of response variability and therefore provides a better overall fit.
2. Model II predicts future observations more accurately because lower coefficients reduce overfitting substantially.
3. Both models possess identical predictive ability because they were fitted to the same experimental dataset.
4. Neither model should be interpreted because regression analysis requires a coefficient of determination equal to one.

Question Number : 86 Question Id : 4410092113399 Question Type : MCQ Option Shuffling : No

A commercially successful rice variety has excellent yield but is highly susceptible to a fungal disease. The breeder wants to improve resistance while preserving most of the original characteristics. Which approach is most suitable?

Options :

1. Induce mutations and select resistant variants with minimal alteration of existing genotype.
2. Increase random mutations until the complete genetic constitution of the crop is replaced.
3. Select only large chromosomal mutations because small mutations lack agricultural importance.
4. Avoid induced mutations because radiation cannot generate disease-resistance characters.

Question Number : 87 Question Id : 4410092111126 Question Type : MCQ Option Shuffling : No

Which term describes a cell's commitment to a specific developmental fate, even if placed in a different environment?

Options :

1. Determination
2. Differentiation
3. Induction
4. Specification

Question Number : 88 Question Id : 4410092111131 Question Type : MCQ Option Shuffling : No

What is the usual ploidy level of the central cell after double fertilization in angiosperms?

Options :

1. Tetraploid
2. Triploid
3. Haploid
4. Diploid

Question Number : 89 Question Id : 4410092111132 Question Type : MCQ Option Shuffling : No

Which pairing of organism and cleavage pattern is correct?

Options :

1. Bird—superficial cleavage
2. Amphibian—rotational cleavage
3. Cephalopod—spiral cleavage
4. Echinoderm—radial cleavage

Question Number : 90 Question Id : 4410092111136 Question Type : MCQ Option Shuffling : No

Which statement best describes the function of homeotic genes in Drosophila development?

Options :

1. They initiate the first zygotic gene expression.
2. They maintain segment boundaries during gastrulation.
3. They regulate mitotic activity in embryonic cells.
4. They specify identity along the anterior-posterior axis.

Question Number : 91 Question Id : 4410092111130 Question Type : MCQ Option Shuffling : No

Match the molecule to its primary role in mammalian fertilization: IZUMO1.

Options :

1. Sperm surface protein required for fusion with the oocyte
2. Zona pellucida glycoprotein binding sperm
3. Zona pellucida protein triggering the acrosome reaction
4. Oocyte membrane protein binding sperm

Question Number : 92 Question Id : 4410092111138 Question Type : MCQ Option Shuffling : No

Which molecule is crucial for specifying ventral neuronal identities in the developing vertebrate neural tube?

Options :

1. BMPs
2. Retinoic Acid
3. Delta-Notch
4. Sonic Hedgehog

Question Number : 93 Question Id : 4410092111140 Question Type : MCQ Option Shuffling : No

Choose the INCORRECT statement regarding sex determination in *Drosophila melanogaster*.

Options :

1. Sxl gene is crucial for sex determination
2. Sex is determined by the number of X chromosomes in each cell.
3. Y chromosome is essential for male development
4. Dosage compensation occurs by upregulating male X chromosome

Question Number : 94 Question Id : 4410092111144 Question Type : MCQ Option Shuffling : No

What is the main function of thrombocytes in mammalian blood?

Options :

1. Fighting infections
2. Transporting oxygen
3. Regulating osmotic pressure
4. Promoting blood clotting

Question Number : 95 Question Id : 4410092111147 Question Type : MCQ Option Shuffling : No

Which immunoglobulin class is most abundant in the serum of healthy adults?

Options :

1. IgM
2. IgA
3. IgG
4. IgE

Question Number : 96 Question Id : 4410092111151 Question Type : MCQ Option Shuffling : No

Which statement about the oxygen-hemoglobin dissociation curve is correct?

Options :

1. A left shift decreases oxygen binding in lungs.
2. A right shift indicates increased affinity for oxygen.
3. The curve is linear across all oxygen pressures.
4. A right shift facilitates oxygen unloading in tissues.

Question Number : 97 Question Id : 4410092111153 Question Type : MCQ Option Shuffling : No

Which division of the peripheral nervous system is responsible for voluntary muscle control?

Options :

1. Parasympathetic nervous system
2. Enteric nervous system
3. Sympathetic nervous system
4. Somatic nervous system

Question Number : 98 Question Id : 4410092111149 Question Type : MCQ Option Shuffling : No

In a patient with severe hemorrhage, which compensatory response is LEAST likely to occur?

Options :

1. Peripheral vasoconstriction
2. Increased heart rate
3. Renin-angiotensin system activation
4. Decreased sympathetic outflow

Question Number : 99 Question Id : 4410092111156 Question Type : MCQ Option Shuffling : No

In which region of the renal tubular system is urea passively reabsorbed to contribute to medullary osmolarity?

Options :

1. Inner medullary collecting duct
2. Proximal convoluted tubule
3. Distal convoluted tubule
4. Thick ascending limb of loop of Henle

Question Number : 100 Question Id : 4410092111159 Question Type : MCQ Option Shuffling : No

Which division of the peripheral nervous system is primarily responsible for initiating non-shivering thermogenesis in mammals?

Options :

1. Enteric nervous system
2. Sympathetic nervous system
3. Parasympathetic nervous system
4. Somatic nervous system